

AI & Data Science Associate Professional JD

As a Data Science Associate Trainee, we seek enthusiastic expertise in Applied Mathematics (includes Probability and Statistics, Linear Algebra, Calculus and Optimization) and robust Programming, Data Structures, and Algorithms understanding, and passion to pursue AI and ML, and skills that involve a robust analytical interpretation, vision for creating efficient and intelligent self-learning applications leveraging Machine learning and Artificial Intelligence.

A Data Science an AI Trainee is responsible for designing and developing complex analytics, data pipelines, and machine learning systems, implementing appropriate ML algorithms, conducting experiments, and staying updated with the latest developments in the field. They work with data to create models, perform statistical analysis, and train and retrain systems to optimize performance. Their goal is to build efficient self-learning applications and contribute to advancements in artificial intelligence.

Objectives of the role

- ☐ **1.** Solve complex problems with multilayered data sets, and optimize existing machine learning libraries and frameworks
- ☐ **2.** Designing and developing
 - ☐ **a.** Machine learning and deep learning systems
 - ☐ **b.** Running machine learning tests and experiments
 - ☐ **c.** Complex Data and Analytics pipelines
- ☐ **3.** Implementing appropriate algorithms, identify differences in data distribution that could potentially affect model performance in real-world applications
- ☐ **4.** Collaborate with data scientists, administrators, data analysts, data engineers, and data architects on production systems and applications
- ☐ **5.** Your ultimate objective will be to create highly efficient self-learning applications that can adapt and evolve over time, pushing the boundaries of AI technology. Join us and be at the forefront of innovation in the field of Data Science and machine learning.

Responsibilities

- ☐ **1.** Serve as data strategist to identify and integrate new datasets that can be leveraged through our product capabilities, Execute analytical experiments to help solve problems across various domains and industries
- ☐ **2.** Study, build and transform data science prototypes and apply appropriate machine learning algorithms and tools
- ☐ **3.** Visualizing data for deeper insights, Studying, transforming, and converting data science prototypes, Verifying data quality and/or ensuring it via data cleaning
- ☐ **4.** Select appropriate datasets and data representation methods, Perform statistical analysis and use results to improve models
- ☐ **5.** Construct optimized data pipelines to feed machine-learning models, Understanding when your findings can be applied to business decisions
 - ☐ **a.** Run machine learning tests and experiments
 - ☐ **b.** Perform statistical analysis and fine-tuning using test results
 - ☐ **c.** Train and retrain systems when necessary
- ☐ **6.** Research and implement appropriate ML algorithms and tools, Extend existing Data Science and machine learning libraries & frameworks
- ☐ **7.** Design machine learning systems, Run machine learning tests and experiments, and document findings and results
- ☐ **8.** Develop machine learning applications according to requirements, extend existing ML libraries, frameworks and keep abreast of developments in the field

Requirements and skills

- ☐ **1.** Proven experience as a Data Science Associate, data mining professional or similar role, 2+ years of expertise applying AI to practical uses, must have skills in pattern recognition and predictive modelling.
- ☐ **2.** Identify valuable data sources and automate collection processes, Undertake preprocessing of structured and unstructured data
- ☐ **3.** Analyze large, complex datasets to extract actionable insights discover trends and patterns and improve machine learning model performance

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Start-up Checklist

- ☐ **4.** Develop and maintain complex data pipelines, including data preprocessing, feature extraction, model training, and evaluation
- ☐ **5.** Impeccable analytical and problem-solving skills and Desire to learn
- ☐ **6.** Extensive math and computer skills, with a deep understanding of probability, statistics, and algorithms
- ☐ **7.** In-depth knowledge of machine learning frameworks, like Keras or PyTorch and libraries (like scikit-learn, NumPy, Pandas, or Keras)
- ☐ **8.** Excellent time management and organizational skills
- ☐ **9.** Familiarity and understanding of data structures, data modelling, and software architecture
 - ☐ **a.** Ability to write robust code in Python
 - ☐ **b.** Adept at using big data processing tools (e.g., Spark) and data storage systems (e.g., SQL, NoSQL)
 - ☐ **c.** Experience with cloud environments, ML Ops
- ☐ **10.** B Sc, B. Tech in Computer Science, Mathematics, or similar field; master's degree in data science, AI/ML is a plus and highly preferred.
- ☐ **11.** B Sc, B. Tech in Computer Science, Mathematics or similar field; master's degree in data science, AI/ML is a plus and highly preferred
 - ☐ **a.** B Sc Computer Science
 - ☐ **b.** B. Tech Computer Science
 - ☐ **c.** B Sc Mathematics/Statistics
 - ☐ **d.** Master's degree in Data Science, AI/ML
 - ☐ **e.** Relevant Internship and 2 Year + Experience
- ☐ **12.** Ability to work in a team and outstanding analytical and critical thinking skills

Selection Process

- ☐ **1. Screening Interviews:** Screening rounds can be conducted before inviting candidates for second-round interviews. These screening interviews can take place virtually and involve an assessment of essential skills, like programming and machine learning, along with a deep dive into the candidate's academic experience, projects, career trajectory, and motivation towards AI.
- ☐ **2. Second-Round Interviews:** The top candidates will be invited to a second interview, either virtually or in person for the series of interviews that will assess the candidate's skills, as described earlier and focused on the core skill set to be evaluated or other critical considerations (*Described in Appendix I*). Assess to get a better understanding of a candidate's approach to problem solving, dealing with ambiguity, code completion tasks and practical skills.
- ☐ **3. Group Discussion:**, Introduce themselves and provide a brief summary of the kind of work they do to each other.
- ☐ **4. Listening the Candidates:** The last five to ten minutes of each interview should be reserved for the candidate to ask questions, critical component, as the types of questions a candidate asks offer a great deal of information about how carefully they've considered the role.
- ☐ **5. Short project (Optional Round):** A short problem statement can be assigned to create a project as short term project with definite timeline of 24-48 hours. Candidate may work on this independently at home and use open resources. *this task offer a tangible demonstration of a candidate's ability to translate theoretical knowledge into applied solutions*

Appendix I

Detailed Subject Breakup	
Topics	Subtopics
Probability and Statistics	⊗ Counting (Permutations and Combinations) ⊗ Probability Axioms ⊗ Sample Space ⊗ Events ⊗ Independent Events ⊗ Mutually Exclusive Events ⊗ Marginal, Conditional, and Joint Probability ⊗ Bayes' Theorem ⊗ Conditional Expectation and Variance ⊗ Mean, Median, Mode, and Standard Deviation ⊗ Correlation and Covariance ⊗ Random Variables ⊗ Discrete Random Variables and Probability Mass Functions (Uniform, Bernoulli, and Binomial Distribution) ⊗ Continuous Random Variables and Probability Distribution Functions (Uniform, Exponential, Poisson, Normal, Standard Normal, t-Distribution, Chi-Squared Distributions) ⊗ Cumulative Distribution Function ⊗ Conditional Probability Density Function ⊗ Central Limit Theorem ⊗ Confidence Interval ⊗ z-Test ⊗ t-Test ⊗ Chi-Squared Test
Linear Algebra	⊗ Vector Space ⊗ Subspaces ⊗ Linear Dependence and Independence of Vectors ⊗ Matrices ⊗ Projection Matrix ⊗ Orthogonal Matrix ⊗ Idempotent Matrix ⊗ Partition Matrix and Their Properties ⊗ Quadratic Forms ⊗ Systems of Linear Equations and Solutions ⊗ Gaussian Elimination ⊗ Eigenvalues and Eigenvectors ⊗ Determinant ⊗ Rank ⊗ Nullity

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Start-up Checklist

	⊗ Projections ⊗ LU Decomposition ⊗ Singular Value Decomposition
Calculus and Optimization	⊗ Functions of a Single Variable ⊗ Limit ⊗ Continuity and Differentiability ⊗ Taylor Series ⊗ Maxima and Minima ⊗ Optimization Involving a Single Variable
Programming, Data Structures, and Algorithms	⊗ Programming in Python ⊗ Basic Data Structures: Stacks, Queues, Linked Lists, Trees, and Hash Tables ⊗ Search Algorithms: Linear Search and Binary Search ⊗ Basic Sorting Algorithms: Selection Sort, Bubble Sort, Insertion Sort ⊗ Divide and Conquer Techniques: Mergesort, Quicksort ⊗ Introduction to Graph Theory ⊗ Basic Graph Algorithms: Traversals and the Shortest Path
Database Management and Warehousing	⊗ ER-Model (Entity-Relationship Model) ⊗ Relational Model: Relational Algebra, Tuple Calculus ⊗ SQL (Structured Query Language) ⊗ Integrity Constraints ⊗ Normal Form ⊗ File Organization ⊗ Indexing ⊗ Data Types ⊗ Data Transformation: Normalization, Discretization, Sampling, and Compression ⊗ Data Warehouse Modelling: Schema for Multidimensional Data Models ⊗ Concept Hierarchies ⊗ Measures: Categorization and Computations
Machine Learning	⊗ Supervised Learning ⊗ Regression and Classification Problems ⊗ Simple Linear Regression ⊗ Multiple Linear Regression ⊗ Ridge Regression ⊗ Logistic Regression ⊗ k-Nearest Neighbours ⊗ Naive Bayes Classifier ⊗ Linear Discriminant Analysis ⊗ Support Vector Machine ⊗ Decision Trees ⊗ Bias-Variance Trade-off ⊗ Cross-validation Methods: Leave-One-Out (LOO) Cross-validation, k-Folds Cross-validation ⊗ Multi-layer Perceptron

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Start-up Checklist

	⊗ Feed-forward Neural Network ⊗ Unsupervised Learning: ⊗ Clustering Algorithms ⊗ k-Means and k-Medoid Clustering ⊗ Hierarchical Clustering ⊗ Dimensionality Reduction ⊗ Principal Component Analysis (PCA)
Artificial Intelligence (AI)	⊗ Search: Informed Search, Uninformed Search, Adversarial Search ⊗ Logic: Propositional Logic, Predicate Logic ⊗ Reasoning under Uncertainty Topics ⊗ Conditional Independence Representation ⊗ Exact Inference through Variable Elimination ⊗ Approximate Inference through Sampling